

Research Topic Meeting Summary — Dr Arnick Abdollahi

Meeting details

- Date: 29 July 2026
- Participants: Dr Arnick Abdollahi and Tuan Nguyen
- Purpose: Explore a feasible Research Project topic and agree on immediate next steps
- Sources compared: Local Whisper transcription and Apple Voice Memos transcription
- Transcript status: Both machine-generated transcripts were compared against each other; decisions are paraphrased because some words and speaker labels remain uncertain

Outcome

Dr Arnick proposed a broad research direction centred on:

Transparent fire-hotspot detection using time-series transformer and/or graph-based predictive models

The exact title, research gap, objectives, questions, and methodology are not final. Tuan must first conduct a focused literature scan to establish what has already been studied and identify a defensible gap.

Proposed technical components

Dr Arnick identified the following potential components:

- Fire-hotspot detection or prediction
- Time-series transformer models
- Graph-based models or graph neural networks
- Possible comparison of transformer and graph-based approaches
- Multi-source data fusion
- Domain-aware feature engineering
- Climate and weather drivers
- Land-use and ground-condition drivers
- Transparent and interpretable modelling
- Actionable insights for decision-makers
- Prediction confidence and uncertainty

These are candidate components, not a requirement to implement everything. The literature gap and available data should determine the final scope.

Research philosophy

Dr Arnick emphasised that the subject should develop research skills rather than repeat a normal coursework assignment. The project should:

- Identify a real-world problem and research gap
- Develop an innovative or defensible methodological contribution
- Progress logically from gap to objectives, research questions, and methodology
- Produce more than a prediction score
- Explain why a location is identified as a fire hotspot
- Communicate confidence, uncertainty, and actionable insight
- Remain feasible within the short subject timeline

Agreed immediate task

Complete a focused literature and gap scan by Friday 31 July or Saturday 1 August 2026.

The scan should investigate:

1. Existing AI and ML methods for fire-hotspot detection or prediction
2. Time-series transformer applications in wildfire or fire-risk modelling
3. Graph-based or graph-neural-network applications in wildfire modelling
4. Multi-source environmental data-fusion methods
5. Common feature-engineering approaches for climate, weather, vegetation, land use, and other drivers
6. Explainability, confidence, calibration, and uncertainty methods
7. Limitations and unresolved gaps that are feasible for one semester

Recommended first-deliverable format

Dr Arnick asked for research into prior work, limitations, and gaps by Friday or Saturday, but he did not prescribe a document length or template. A practical short deliverable could contain:

- Working problem statement
- Search strategy and keywords
- Literature comparison table
- Summary of what has already been done
- Recurring limitations
- Two or three candidate research gaps
- Two or three possible project formulations
- Initial data requirements
- Feasibility and risk notes
- Recommended direction for discussion

Do not finalise the objectives, research questions, or methodology before identifying the gap.

Administrative action

Dr Arnick advised emailing Dr Tony Huang immediately to confirm the preferred supervisor and broad topic. The final title is not required yet. His wording indicates willingness to work together if Tuan confirms this choice; the formal administrative appointment should therefore be treated as pending confirmation.

Dr Huang's Canvas announcement confirms that students must have both a research topic and supervisor confirmed in order to enrol in the subject.

The confirmation should state:

- Dr Arnick Abdollahi is the proposed supervisor
- The broad direction is transformer-based and/or graph-based modelling for climate-hazard or fire-hotspot detection
- The exact research question will be refined after the initial literature and gap analysis

Dr Arnick should be copied on the confirmation email.

Supervisor and broad-topic confirmation email

Dr Arnick referred to emailing "Tony." The Canvas announcement confirms the relevant academic is Dr Tony Huang.

To: weidong.huang@uts.edu.au

CC: arnick.abdollahi@uts.edu.au

Dear Dr Huang,

I am writing to confirm my proposed supervisor and broad topic direction for 36126 Innovation Lab: Research Project.

Following our discussion today, I would like to confirm Dr Arnick Abdollahi as my preferred supervisor. Dr Abdollahi is copied on this email. Our current broad direction is transparent predictive modelling for climate hazards, focusing on fire-hotspot detection using time-series transformer and/or graph-based models, with consideration of multi-source data fusion, feature engineering, explainability, confidence, and uncertainty.

The exact title, research gap, objectives, and methodology will be refined with Dr Abdollahi after my initial literature and gap analysis.

Could you please confirm whether any additional form, approval, or enrolment step is required?

Kind regards,
Tuan Nguyen
Student ID: [add student ID]
Master of Data Science and Innovation
manhtuan.nguyen@student.uts.edu.au

Time commitment

Tuan indicated an expected commitment of approximately 12 hours per week or more. Dr Arnick recommended:

- Regular weekly planning
- Dividing the project into explicit tasks
- Making consistent progress every week
- Managing the short timeline carefully

Timeline information

- Dr Arnick believed the final subject deadline may be near the end of October, but this was not confirmed.
- Dr Huang stated that detailed assessment requirements and the presentation venue would be released within one or two weeks.
- The authoritative submission and presentation dates must still be checked in Canvas.

Confirmed subject requirements from Canvas

Dr Tony Huang's welcome announcement establishes the following:

- The subject is a self-guided research project supported by an academic supervisor.
- A confirmed topic and supervisor are required for enrolment.
- The work must address research questions grounded in relevant academic literature.
- The project must produce an original contribution: a new or improved method, or new insights derived from data.
- Merely applying existing statistical or machine-learning methods to a dataset is insufficient.
- Supervisors are asked to provide at least 12 contact hours during the semester; the supervision arrangement should be negotiated directly.
- Projects requiring high-risk ethics approval or external confidentiality agreements are not recommended unless already cleared by the supervisor.
- The final research report contributes 50% of the final mark, informed by supervisor feedback.
- A face-to-face end-of-semester presentation contributes the remaining 50%.
- Detailed assessment and venue information will be published separately.

Official study guide: [What is Research?](<https://studyguides.lib.uts.edu.au/36126/whatisresearch>)

Communication

- Use Microsoft Teams to send questions, ideas, drafts, and progress updates.
- Continue refining the project through discussion with Dr Arnick.

Decisions

Item	Decision
Preferred supervisor	Dr Arnick Abdollahi
Broad domain	Climate hazard / wildfire AI
Working task	Fire-hotspot detection or prediction
Candidate methods	Time-series transformers and/or graph-based models
Essential research qualities	Data fusion, feature engineering, transparency, actionable insights, confidence, uncertainty
Immediate deliverable	Focused literature and research-gap scan
Deliverable target	31 July or 1 August 2026
Weekly commitment	Approximately 12+ hours
Communication	Microsoft Teams
Formal confirmation	Email coordinator now; copy Dr Arnick

Transcript comparison notes

The Apple and Whisper transcripts agree on the substantive decisions:

- Confirm the preferred supervisor and broad direction with Tony now, copying Dr Arnick.
- Begin with a literature scan of prior work, limitations, and defensible gaps.
- Send the initial findings by Friday or Saturday.
- Use “fire hotspot detection” as the initial search phrase.
- Investigate transformer-based and graph-based modelling, including the possibility of comparing them.
- Consider data fusion, domain feature engineering, transparency, confidence, and uncertainty.
- Define objectives, research questions, and methodology only after identifying the gap.
- Use Microsoft Teams for ongoing supervision communication.

The Apple transcript provides two useful clarifications:

1. Dr Arnick did not require both model families. One may be selected, or a comparison may be considered if feasible.
2. He did not specify a formal page count or template for the first deliverable.

The transcripts do not resolve the final deadline, exact geographic scope, available datasets, prediction horizon, or required first-deliverable format.

Open questions

- Does confirming the topic and supervisor by email complete enrolment, or is an additional administrative step required?
- What is the authoritative final assessment deadline?
- What are the report requirements, presentation date, and presentation venue?
- Should the first literature scan remain geographically broad or focus on Australia?

- Is a concise literature matrix plus two or three candidate gaps suitable for the first deliverable?
- Which datasets are already available through Dr Arnick?
- Is the target fire occurrence, hotspot classification, susceptibility, or future risk?
- What are the geographic scope, spatial unit, and forecast horizon?
- Should the final study compare transformers and graph models, or select only one?
- What baseline models are required?
- What definition of transparency and actionable insight will be evaluated?
- Which uncertainty or calibration method is appropriate?
- What compute and storage resources are available?
- When is the next supervision meeting?
- How will the minimum 12 supervisor-contact hours be scheduled and recorded?

Two-day action plan

29 July

- Send supervisor/topic confirmation to the coordinator and copy Dr Arnick.
- Create the literature-search protocol.
- Define inclusion and exclusion criteria.
- Run initial searches across scholarly databases.
- Collect foundational and recent papers.

30 July

- Screen titles and abstracts.
- Populate the literature comparison matrix.
- Separate transformer, graph-based, data-fusion, and explainability studies.
- Record datasets, targets, spatial/temporal units, validation methods, and limitations.

31 July

- Consolidate recurring gaps.
- Draft two or three feasible research formulations.
- Evaluate each against data readiness, novelty, compute, and semester scope.
- Write the short deliverable.

1 August, contingency

- Quality-check citations and claims.
- Revise the recommended direction.
- Send the deliverable to Dr Arnick through Teams.
- Ask for feedback and the next meeting time.

Initial search terms

Use combinations of:

- "wildfire hotspot detection" machine learning
- "fire hotspot prediction" time series
- wildfire "time series transformer"
- wildfire "temporal fusion transformer"
- wildfire "graph neural network"
- "fire risk" GNN spatiotemporal

- wildfire multisource "data fusion"
- wildfire prediction SHAP explainable AI
- wildfire prediction uncertainty calibration
- wildfire occurrence remote sensing climate land use
- bushfire hotspot prediction Australia machine learning

Literature matrix fields

Field	What to record
Citation	Authors, year, title, venue
Research task	Detection, occurrence, susceptibility, risk, spread, or severity
Geography	Country/region and spatial unit
Time scale	Daily, weekly, monthly, seasonal, or event-based
Data	Satellite, climate, weather, vegetation, land use, topography, human factors
Fusion approach	Early, intermediate, late, graph-based, or other
Model	Baseline and advanced models
Validation	Random, temporal, spatial, or spatiotemporal split
Metrics	Classification, calibration, spatial, or decision metrics
Explainability	SHAP, feature importance, attention, counterfactuals, or none
Uncertainty	Calibration, ensemble, Bayesian, conformal, or none
Main finding	Supported result
Limitation	Author-stated and independently observed
Relevance	How it informs the proposed project

Candidate formulations to test against the literature

These are hypotheses for exploration, not final topics:

1. Transformer direction: Determine whether a time-series transformer improves fire-hotspot prediction over interpretable tree-based and recurrent baselines under temporal and spatial holdouts.
2. Graph direction: Represent spatial regions and their environmental relationships as a graph, then evaluate whether a spatiotemporal graph model improves hotspot prediction and regional generalisation.
3. Transparency direction: Compare whether region-specific explanations, calibrated confidence, and uncertainty estimates provide more reliable actionable insight than a single black-box prediction model.
4. Fusion direction: Evaluate which fusion of climate, weather, vegetation, land-use, and spatial-context features produces robust gains without introducing leakage.

The final project should likely combine one primary modelling contribution with transparency and robust validation, rather than attempting all four directions.

Recommended scope discipline

A feasible project may use:

- One clearly defined target
- One geographic resolution
- One prediction horizon

- Two or three simple baselines
- One advanced model family
- One explainability approach
- One uncertainty or calibration analysis
- Spatial and temporal validation
- A concise decision-oriented interpretation

Avoid committing to both a full transformer study and a full GNN study until the literature, data, and compute requirements are understood.